

OSBA Route Study Overview

This revised route study uses increased ridership counts for public and non-public routes to specifically illustrate the impact on transportation services. Areas examined were:

- Future ridership for High School and Middle School buildings
- Future ridership for grades Kindergarten through 5th grade
- Future ridership for pre-school and non-public schools
- Public and non-public school buildings bell schedules
- Alternative routing options
- Estimated cost of each routing option

Note that the routing models developed are reflective of the enforcement of Board Policy 8600, one mile walking limits at all grade levels.

Determining Ridership

In any routing study, the number of students that ride the buses must be identified first. BBHCSD staff has estimated that 93% of average daily membership (ADM) in grades Kindergarten through grade 5 will be riding the bus equaling 1,514 riders.

Ridership numbers for Middle School students are estimated to be 825 or 92% of the ADM. High school ridership numbers are estimated to be 807 or 60% of the ADM.

BBHCSD has estimated the number of non-public schools' riders to be 298 riders. BBHCSD has estimated more than 30 pre-school riders, who will be transported on dedicated buses.

Ridership Data	ADM	Bus Riders
Elementary School	1628	1514
Preschool	75	30
Middle School	955	860
High School	1382	829
HS/MS Combined Total	2337	1689
Assumption Academy	55	10
Old Trail School	36	17
St. Alberts	91	55
St. Michaels	41	23

Historically, BBHHS ridership varies significantly. As High School students are very involved in before and after school activities throughout the year, ridership drops. It is my opinion that the High School ridership number may be somewhat inflated.

Plan D-1: **Note that the routing model developed uses the current bell schedule that is in place at our elementary schools. OSBA (Doug Palmer) has been requested to update the routing structure accordingly. This will not reduce the number of buses needed to support the elementary school but will change the amount of time being shown between the hypothetical bell schedules used to construct this routing model.**

Uses 48 buses, in a modified two-tier scenario with the Middle School and High School students transported on the first tier, elementary students on the second tier as follows:

- Morning and afternoon transportation
- 34 buses transport middle and high school students, six buses will transport students with special needs to school.
- 34 buses transport elementary school students to school and six buses will transport students with special needs to school.
- Seven buses transport non-public students.
- One bus will transport students to Cuyahoga Valley Career Center (CVCC)

The total bus count identified is in line with two tier routing models developed internally. It is extremely difficult to identify/structure special needs routing with confidence due to the volatility of the student population's transportation needs.

Plan E-1: **This was initially the OSBA preferred routing structure.**

In this plan, we continue the option of a three-tier scenario using 39 buses. The high school students are transported on the first tier, middle school students on the second tier and the elementary students are transported on the last tier. The routing plan is as follows:

Morning and afternoon transportation

- 25 buses transport high school students on the first tier, with an average of 37 students assigned to each bus.
- 23 buses transport middle school students on the second tier while 7 buses transport non-public.
- One bus will transport students to CVCC.
- 33 Buses will be used to transport the elementary students.
- Six buses transport students with special needs throughout the tiers including the pre-school students.

The three tier model developed provides such significant time between tiers that it would result in the High School students starting even earlier and elementary school students starting even later.

With 45 to 50 minutes between bell times for the High School and Middle it is not that dissimilar to our current routing structure with the exception that the Elementary School would most likely need an even longer time after the Middle School bells to perform the third tier of transport.

Cost Analysis

The cost impact associated with each routing plan is prepared by using the operating cost of each bus as reported on the FY2020 ODE T2.

- 1) Implementing Plan D, (elementary first tier) a two-tier routing plan using 48 buses, has an estimated first year cost of \$3,947,037. The operating cost will continue to be \$2,978,037 annually.
- 2) Implementing Plan D-1, (elementary last tier) a two-tier routing plan using 48 buses, has an estimated first year cost of \$3,996,957. The operating cost will continue to be \$3,027,957 annually.
- 3) Implementing Plan E-1, (elementary/high/middle tiers) a three-tier routing plan using 39 buses, has an estimated operational cost of \$2,527,915 annually.
- 4) Implementing Plan E-1, (high/middle/elementary tiers) a three-tier routing plan using 39 buses, has an estimated operational cost of \$2,654,060 annually.

The three tier reduced operating expense identified is directly related to nine fewer buses being identified as needed to support the three tier routing model. The cost structure developed by OSBA is support of the three tier routing model does not explain the impact of the Elementary School. Again, the cost savings identified is directly related to the suggestion that fewer buses are needed to support the three tier routing model.

With the length of day implications resulting from a three tier routing structure it is likely that any savings generated by the three tier model would be offset by building level expenses needed to manage students beyond the teacher work day.

The estimated annual operating cost impacts do not reflect the capitalization cost of future bus purchases. Bus purchase plans with a 60-bus fleet, 48-route buses and 12 spares should have a goal to replace 7 buses annually to maintain a 10-year replacement cycle and will have an estimated cost of \$710,000 each year.

We have budgeted to purchase three (3) buses per year for the next three years. We will monitor not just the “average age” of the fleet but the condition of our buses to determine our number of buses proposed for purchase each year.

3327.01 Transportation of Pupils:

Ohio statute has changed and will be effective next school year. If the school district chooses to take advantage of this change and consider our private/parochial students being dropped off up to 30 minute prior to their scheduled school start time and picked up no later than 30 minutes after the close of the school it provide the flexibility to structure shuttle to these schools.

<http://education.ohio.gov/Topics/Finance-and-Funding/School-Transportation/New-pupil-transportation-provisions-in-Ohio-s-budg>